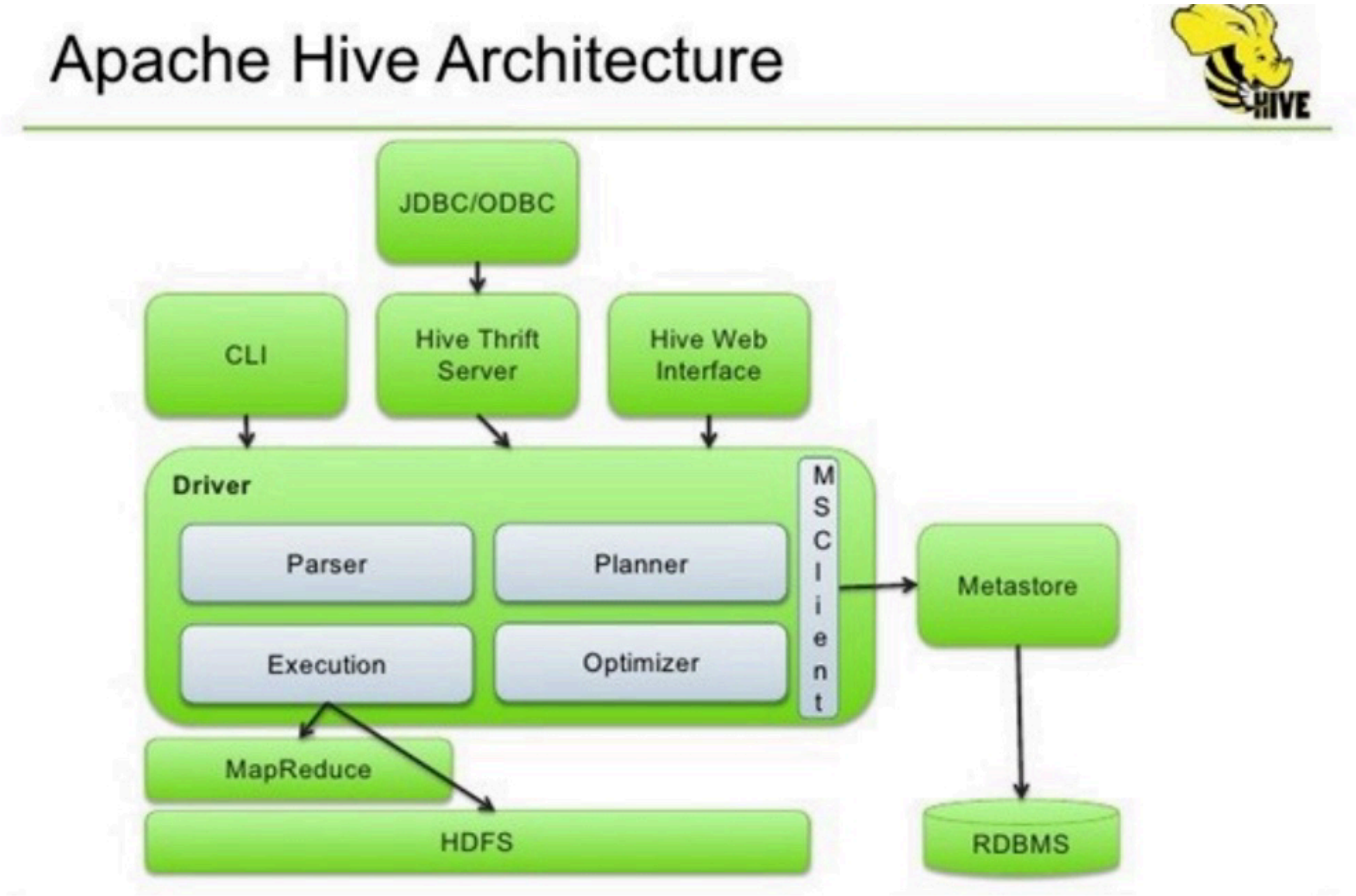
Store Data Effieciently in Big Data

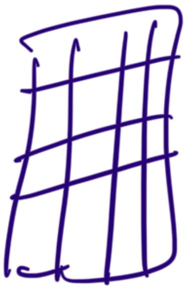
Hive architecture

How does Apache hive act as a data warehouse?
What makes hive as Hive? (Architecture)
How to interact with Hive?
What are different HQL commands that are used commonly?
What are the optimization techniques in Hive ?
Use cases of hive/ where to use hive/why and why not
File Formats



Receive SQL query

1. Parse HiveQL
2. Make optimizations
3. Plan execution
4. Submit job(s) to cluster
5. Monitor progress
6. Process data in MapReduce or Apache Spark
7. Store the data in HDFS

Hive 

Metastore ⇒ Metadata
↳ Column name
↳ Column datatypes

+

HDFS ⇒ Actual Data

Execution flow

1. Write a sql query & submit
2. Parse Hive QL
3. * Make optimizations
4. plan execution (HQL ↓ Jar)
5. Submit jobs to cluster
6. Monitor Progress
7. Process data in MapReduce
8. Store the output to HDFS

① Hive Metastore

↳ is a service to store metadata for Hive tables.

↳ A table in Hive, it will be represented as directory in HDFS.

/user/hive/warehouse → default directory

SQL → namespace/database/keyspace

Aman → Manish | Rajaram → Google

employee | employee

Select * from Manish.employee;

↳ metadata inside a RDBMS

mysql

Question : "In 2019,Get the details of all travelers about their choice of stays in New York city i.e whether they book Entire home/apt or Private Room or they preferred Shared room"

An. 1 NYC - 2019. csv
Hive

1. Create database ✓
2. Create table ✓
3. Load the data ✓
4. Verify if data is in properly loaded or not ✓
5. query
6. Show the output

1. DDL → Data Definition Language

1. Create Database / table / view
Procedure / function
2. Drop
3. Truncate
4. Alter
5. Show → metadata of DB, tables
6. ⇒ Describe → detail information about tables, DB

Languages

② DML : Data manipulation –

① Load
② Insert
③ Update
④ Delete
⑤ Export
⑥ Import

File format

↳ Text files
↳ Basic & human readable files
↳ CSV/TSV/xlsx

① Consumes more space to store numeric value as string.

↳ Sequence File format :-
↳ Stores data in a key value pair in a

binary format.

- ↳ More efficient than a text file
- ↳ Not human readable

↳ Avro File format
- ↳ embeds Schema metadata in file (JSON)
- ↳ Ideal for long term storage
- ↳ Efficient storage
- ↳ widely supported across Hadoop ecosystem
- ↳ learning curve is slightly higher side.

.avro

avro_tools

(4) Parquet file format
- ↳ efficient among all
- ↳ columnar format
- ↳ very fast for massive read & write ops
- ↳ Not human readable

(5) ORC file
storage

↳ easy to use

↳ open source

↳ easy to work with

↳ highly optimize for Big Data use cases.

```
{
  "type": "record",
  "name": "Employee",
  "fields": [
    {"name": "id", "type": "int"},
    {"name": "name", "type": "string"},
    {"name": "middle_name", "type": ["null", "string"], "default": null},
    {"name": "salary", "type": ["null", "double"], "default": null}
  ]
}
```

{"name" : "middle_nam", type , "size" = 32)